```
# from google.colab import files
# upload = files.upload()

# upload.keys()

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
# Load dataset
df = pd.read_csv('Sample - Superstore (1).csv', encoding = 'latin1')

# Basic Analysis

# Q1 Check first five columns, no.of rows and cols, data type, null values, duplicate values

df.head()

df.describe()

df.columns.value_counts().sum()

df.shape

df.dtypes

df.isnull().sum()

df.duplicated().sum()
```

```
np.int64(0)
```

```
# Intermediate Analysis

# Q2 Total sales
df['Sales'].sum()

# Q3 Total profit
df['Profit'].sum()

# Q3 Average discount
df['Discount'].mean()

# Q4 Total orders
df['Order ID'].nunique()

# Q5 Number of unique cities, customers, categories
df['City'].nunique()

df['Customer Name'].nunique()

df['Category'].nunique()
```

```
3
```

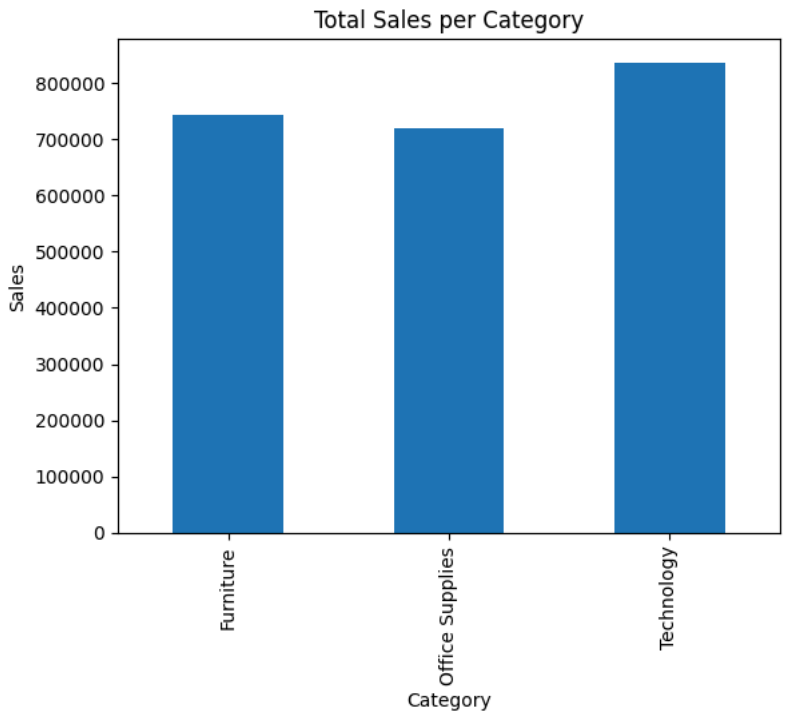
```
# Advance Analysis

# Q6 Total sales per category
df.groupby('Category')['Sales'].sum()

# Q7 Total profit per category
df.groupby('Category')['Profit'].sum()

# Q8 Top 10 products by sales
df.groupby('Product Name')['Sales'].sum().sort_values(ascending = False).head(10)

# Q9 Top 10 products by profit
df.groupby('Product Name')['Profit'].sum().sort_values(ascending = False).head(10)

# Q10 Top 10 cities with highest sales
df.groupby('City')['Sales'].sum().sort_values(ascending = False).head(10)
```

```
# Q11 Top 10 cities with lowest sales
df.groupby('City')['Sales'].sum().sort_values(ascending = True).head(10)

# Q12 Top 10 region by sales
df.groupby('Region')['Sales'].sum().sort_values(ascending = False).head(10)

# Q13 Top 10 region by profit
df.groupby('Region')['Profit'].sum().sort_values(ascending = False).head(10)

# Q14 Top ship mode
df['Ship Mode'].value_counts().head(1)

# Q15 Products with highest discount
df[df['Discount'] == df['Discount'].max()]
```

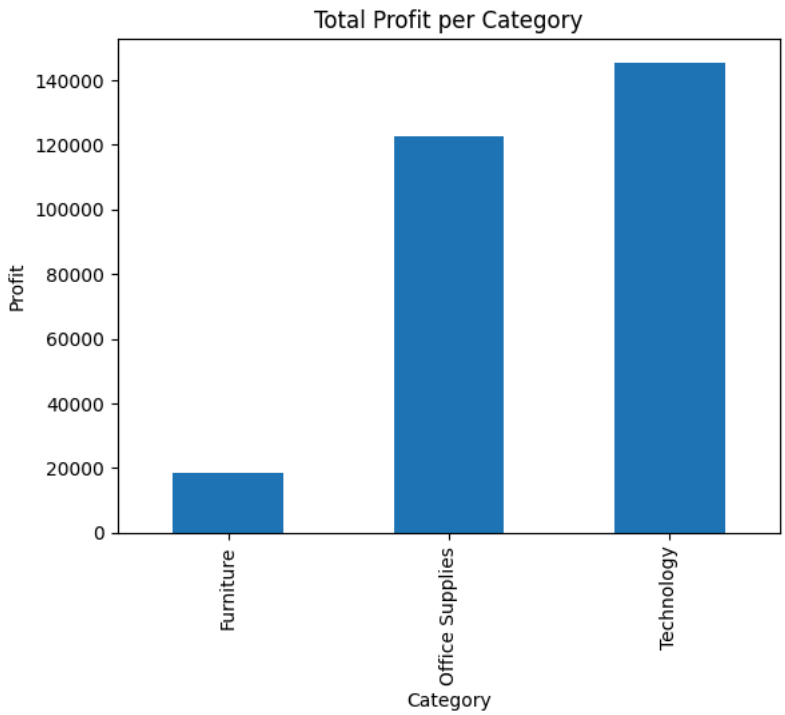
	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	...	Postal Code	Region	Produc I
14	15	US-2015-118983	11/22/2015	11/26/2015	Standard Class	HP-14815	Harold Pawlan	Home Office	United States	Fort Worth	...	76106	Central	OFF-AF 1000231
15	16	US-2015-118983	11/22/2015	11/26/2015	Standard Class	HP-14815	Harold Pawlan	Home Office	United States	Fort Worth	...	76106	Central	OFF-B 1000075
75	76	US-2017-118038	12/9/2017	12/11/2017	First Class	KB-16600	Ken Brennan	Corporate	United States	Houston	...	77041	Central	OFF-B 1000418
101	102	CA-2016-158568	8/29/2016	9/2/2016	Standard Class	RB-19465	Rick Bensley	Home Office	United States	Chicago	...	60610	Central	OFF-B 1000260
169	170	CA-2014-139892	9/8/2014	9/12/2014	Standard Class	BM-11140	Becky Martin	Consumer	United States	San Antonio	...	78207	Central	OFF-AF 1000251
...	...	...	...	...	...	...	...	...	...	...	...	...	...	.
9794	9795	CA-2014-127166	5/21/2014	5/23/2014	Second Class	KH-16360	Katherine Hughes	Consumer	United States	Houston	...	77070	Central	OFF-B 1000097
9795	9796	CA-2016-125920	5/21/2016	5/28/2016	Standard Class	SH-19975	Sally Hughsby	Corporate	United States	Chicago	...	60610	Central	OFF-B 1000342
9808	9809	CA-2017-145093	7/21/2017	7/26/2017	Standard Class	PT-19090	Pete Takahito	Consumer	United States	Chicago	...	60623	Central	OFF-B 1000111
9835	9836	CA-2016-126627	10/10/2016	10/12/2016	First Class	WB-21850	William Brown	Consumer	United States	La Porte	...	77571	Central	OFF-B 1000159
9920	9921	CA-2016-149272	3/15/2016	3/19/2016	Standard Class	MY-18295	Muhammed Yedwab	Corporate	United States	Bryan	...	77803	Central	OFF-B 1000423

300 rows × 21 columns

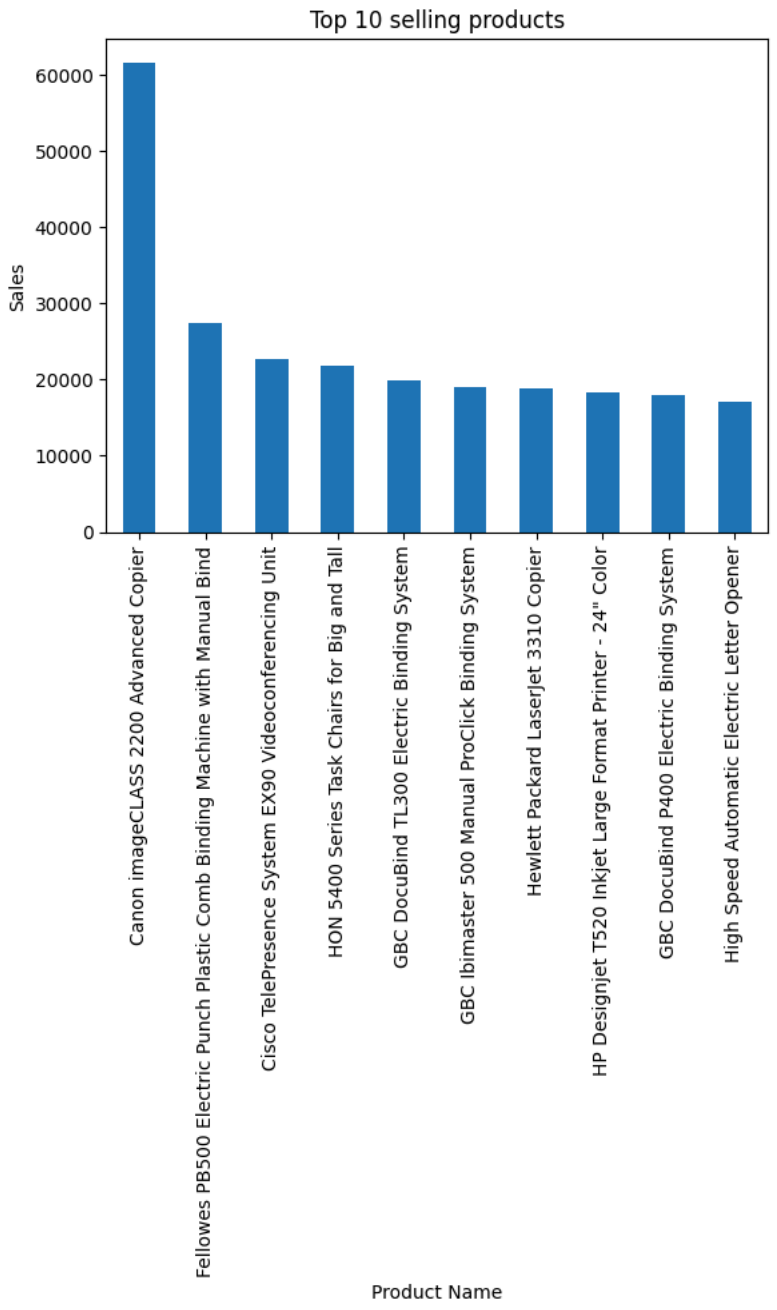
```
# Data visualization

# Q1 Total sales per category

sales_per_cat = df.groupby('Category')['Sales'].sum()
plt.figure()
sales_per_cat.plot(kind = 'bar')
plt.xlabel('Category')
plt.ylabel('Sales')
plt.title('Total Sales per Category')
plt.show()
```



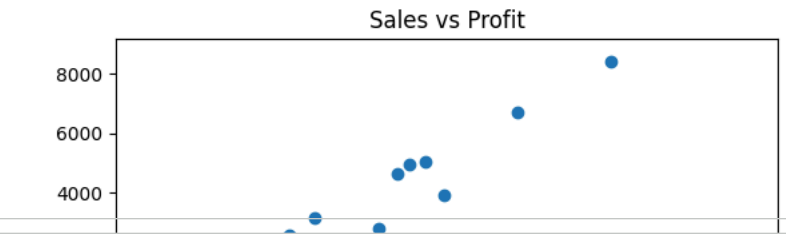

```
# Q2 Total profit per category

sales_per_cat = df.groupby('Category')['Profit'].sum()
plt.figure()
sales_per_cat.plot(kind = 'bar')
plt.xlabel('Category')
plt.ylabel('Profit')
plt.title('Total Profit per Category')
plt.show()
```

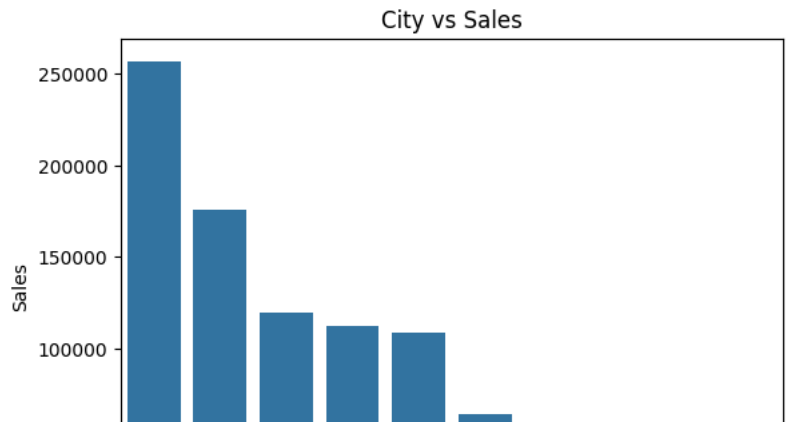



```
# Q3 Top 10 products per sales

top10_products = df.groupby('Product Name')['Sales'].sum().sort_values(ascending = False).head(10)
plt.figure()
top10_products.plot(kind = 'bar')
plt.xlabel('Product Name')
plt.ylabel('Sales')
plt.title('Top 10 selling products')
plt.show()
```




```
#Q4 Top 10 products with highest profits

plt.scatter(df['Sales'], df['Profit'])
plt.xlabel('Sales')
plt.ylabel('Profit')
plt.title('Sales vs Profit')
plt.show()
```




```
#Q5 Top 10 cities with highest sales

top10_cities = df.groupby('City')['Sales'].sum().sort_values(ascending = False).head(10).reset_index()
plt.figure()
sns.barplot(data = top10_cities, x = 'City', y = 'Sales')
plt.xlabel('City')
plt.ylabel('Sales')
plt.title('City vs Sales')
plt.xticks(rotation = 90)
plt.show()
```